Department of Meteorology

Ekman Spirals with Variable Eddy Viscosity

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Contents

1 Background

1.1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and current in the deep ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is often at an angle from the flow aloft (or down under in the ocean) by an angle of more or less 45 ° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft (or current below). This is known as the Ekman transport and is crucial for studying both meteorology and oceanography.

However, this derivation with a surface angle of 45 ° assumes a constant viscosity in the medium. However, this is a simplification of reality, where a turbulent boundary layer is present. Thus, in reality the viscosity should vary with altitude. This would then vary the surface angle from 45 °, and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done both theoretically and numerically and will also be compared to model data. This will give a deeper insight into the Ekman spiral and provide insight into when a surface angle different from 45 ° can be observed.

2 Theory

2.1 Problem formultation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \dfrac{\partial}{\partial z}\left(\nu\dfrac{\partial u}{\partial z}\right), & \text{(1a)} \\ fu = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial y} + \dfrac{\partial}{\partial z}\left(\nu\dfrac{\partial v}{\partial z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will thus approach geostrophic balance, $(u, v) \to (u_g, v_g)$, as were one can neglect the friction term in Equation 1b. Subtracting the geostrophic balance from Equation 1b thus gives

$$\begin{cases} -f(v - v_g) = \dfrac{\partial}{\partial z}\left(\nu\dfrac{\partial u}{\partial z}\right), & \text{(2a)} \\ f(u - u_g) = \dfrac{\partial}{\partial z}\left(\nu\dfrac{\partial v}{\partial z}\right). & \text{(2b)} \end{cases}$$

This problem can be formulated with one equation if one introduces a complex velocity defined as

$$U \equiv u + iv, \tag{3}$$

and a complex geostrophic velocity as $U_g = u_g + iv_g$. Using Equation 2b with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\boxed{\frac{\partial}{\partial z}\left(\nu\frac{\partial U}{\partial z}\right) = if(U - U_g).} \tag{4}$$

With the boundary conditions

$$U(0) = 0, \tag{5a}$$
$$U(z \to \infty) = U_g. \tag{5b}$$

2.2 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude, ν = constant, and can be found in Ekman [1905]. In this case he classical Ekman spiral solution from Equation 4 can obtained as

$$U(z) = U_g \left(1 - e^{-\frac{1+i}{\sqrt{2}} \frac{z}{h_{\text{Ek}}}}\right), \tag{6}$$

where

$$h_{\text{Ek}} \equiv \sqrt{\frac{\nu}{f}} \tag{7}$$

is the Ekman depth scale. One can note that by taking the argument of the change by altitude $\frac{\partial U}{\partial z}|_{z=0}$, one can measure the angle of Equation 6 as $45\,^\circ$.

2.3 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. One can label the lower level as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption one would obtain a superimposed solution similar to Equation 6 as

$$\begin{cases} U_1(z) = A_1 \left(1 - e^{\frac{1+i}{\sqrt{2}} \frac{z}{h_{\text{Ek1}}}}\right) + B_1 \left(1 - e^{-\frac{1+i}{\sqrt{2}} \frac{z}{h_{\text{Ek1}}}}\right), & \text{(8a)} \\ U_2(z) = B_2 \left(1 - e^{-\frac{1+i}{\sqrt{2}} \frac{z}{h_{\text{Ek2}}}}\right) + U_g & \text{(8b)} \end{cases}$$

where the Ekman depth scales are defined for each layer like in Equation 7. To solve for the constants above one has to use the fact that the material flux has to be constant. This gives the jump condition as

$$U_1 = U_2 \quad \text{at } z = H, \tag{9a}$$
$$\nu_1 \frac{\partial U_1}{\partial z} = \nu_2 \frac{\partial U_2}{\partial z} \quad \text{at } z = H. \tag{9b}$$

Using this and the boundary conditions at Equation 5 one obtains a relation between the factors as

$$A_1 - B_1 = U_g \frac{e^{\sqrt{2}(1+i)\varphi} - R}{e^{\sqrt{2}(1+i)\varphi} - R} \tag{10}$$

where φ is the non-dimensional height defined as

$$\varphi = \frac{H}{h_{\text{EK1}}}. \tag{11}$$

The constant R is defined as

$$R = \frac{1 - \frac{\nu_1}{\nu_2}}{1 + \frac{\nu_1}{\nu_2}}. \tag{12}$$

Using this one can define the surface angle as

$$\theta = \arg \left.\frac{\partial U_1}{\partial z}\right|_{z=0} = 45^\circ + \arctan\left[\frac{2Re^{\sqrt{2}\,\varphi} \sin(\sqrt{2}\,\varphi)}{e^{2\sqrt{2}\,\varphi} - R^2}\right] \tag{13}$$

2.4 The exponential decaying viscosity

2.5 The 1.5 layer model

2.5.1 Simple viscosity pofiles for the 1.5 layer model

3 Methodology

4 Results

4.1 The theoretical models

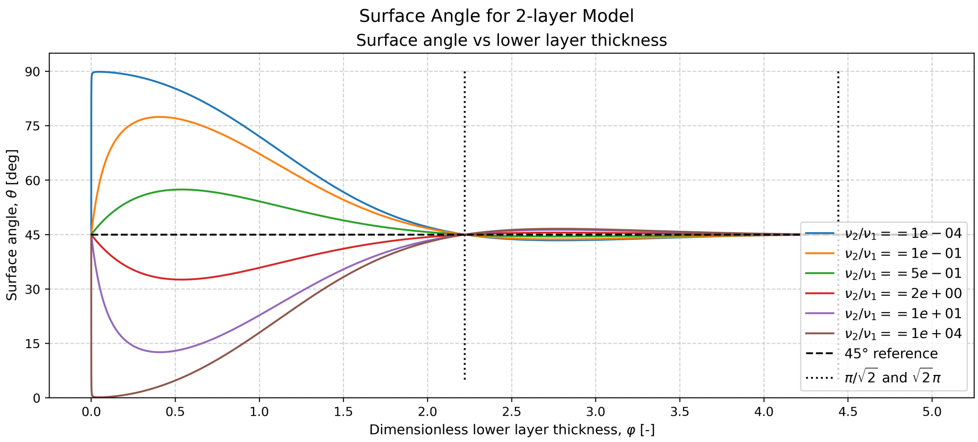

Figure 1: Surface angle as a function of the dimensionless layer thickness $\varphi = H/h_1$ for the two-layer viscosity model. Different curves correspond to different viscosity ratios $\epsilon = h_2/h_1$. The classical Ekman angle of 45° is shown as a reference.

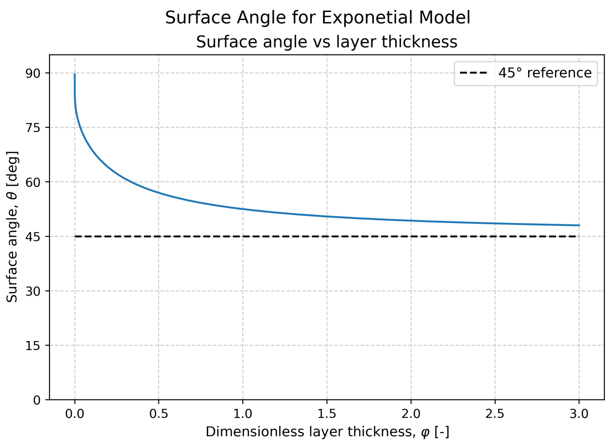

Figure 2: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter φ. The figure illustrates how vertical variations in eddy viscosity modify the classical Ekman turning angle.

4.2 The 1.5 model

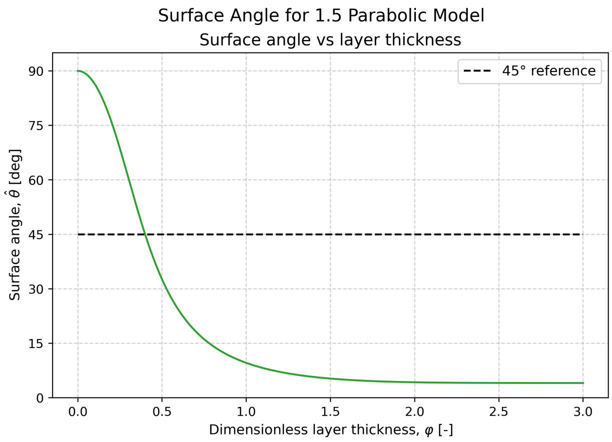

Figure 3: Numerically computed surface angle for the parabolic viscosity profile in the 1.5-layer model. The angle is shown as a function of the normalized layer thickness for different maximum viscosities.

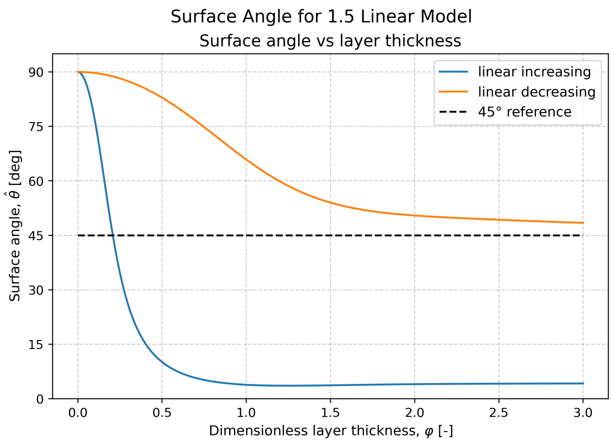

Figure 4: Surface angle obtained numerically for linearly increasing and decreasing viscosity profiles in the 1.5-layer model. The results demonstrate how the vertical structure of viscosity affects the near-surface wind turning.

5 Discussion

6 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

References

V. Walfrid Ekman. On the Influence of the Earth's Rotation on Ocean-Currents. *Arkiv för matematik, astronomi och fysik*, 2(11):1–52, May 1905.